

Periodic Table

1 H 1.008																	2 He 4.003
3 Li 6.941	4 Be 9.012											5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18
11 Na 22.99	12 Mg 24.31											13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.06	17 Cl 35.45	18 Ar 39.95
19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.88	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.38	31 Ga 69.72	32 Ge 72.59	33 As 74.92	34 Se 78.96	35 Br 79.90	36 Kr 83.80
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.94	43 Tc 101.1	44 Ru 101.1	45 Rh 102.9	46 Pd 106.4	47 Ag 107.9	48 Cd 112.4	49 In 114.8	50 Sn 118.7	51 Sb 121.8	52 Te 127.6	53 I 126.9	54 Xe 131.3
55 Cs 132.9	56 Ba 137.3	57 La 138.9	72 Hf 178.5	73 Ta 180.9	74 W 183.9	75 Re 186.2	76 Os 190.2	77 Ir 192.2	78 Pt 195.1	79 Au 197.0	80 Hg 200.6	81 Tl 204.4	82 Pb 207.2	83 Bi 209.0	84 Po	85 At	86 Rn
87 Fr	88 Ra 226.0	89 Ac	104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt									

* Lanthanide Series

** Actinide Series

58 Ce 140.1	59 Pr 140.9	60 Nd 144.2	61 Pm	62 Sm 150.4	63 Eu 152.0	64 Gd 157.3	65 Tb 158.9	66 Dy 162.5	67 Ho 164.9	68 Er 167.3	69 Tm 168.9	70 Yb 173.0	71 Lu 175.0
90 Th 232.0	91 Pa	92 U 238.0	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr

Solubility rules for ionic solids in water

Soluble in water

Soluble	Exceptions	
	Insoluble	Slightly soluble
Most chlorides	AgCl, Hg ₂ Cl ₂	PbCl ₂
Most bromides	AgBr, Hg ₂ Br ₂ , HgBr ₂	PbBr ₂
Most iodides	AgI, Hg ₂ I ₂ , HgI ₂ , PbI ₂	
All nitrates	No exceptions	
Most sulfates	SrSO ₄ , BaSO ₄ , HgSO ₄ , PbSO ₄	CaSO ₄ , Ag ₂ SO ₄

Insoluble in water

Insoluble	Exceptions	
	Soluble	Slightly soluble
Most hydroxides	NaOH, KOH, Ba(OH) ₂ (NH ₄ OH does not exist)	Ca(OH) ₂ , Sr(OH) ₂
Most carbonates	Na ₂ CO ₃ , K ₂ CO ₃ , (NH ₄) ₂ CO ₃	
Most phosphates	Na ₃ PO ₄ , K ₃ PO ₄ , (NH ₄) ₃ PO ₄	
Most sulfides	Na ₂ S, K ₂ S, (NH ₄) ₂ S	

Standard Reduction Potentials at 25°C

Half-reaction	E°(volts)
F ₂ (g) + 2 e ⁻ ⇌ 2 F ⁻ (aq)	+ 2.87
H ₂ O ₂ (aq) + 2 H ⁺ (aq) + 2 e ⁻ ⇌ 2 H ₂ O(l)	+ 1.78
PbO ₂ (s) + SO ₄ ²⁻ (aq) + 4 H ⁺ (aq) + 2 e ⁻ ⇌ PbSO ₄ (s) + 2 H ₂ O(l)	+ 1.69
2 HClO(aq) + 2 H ⁺ (aq) + 2 e ⁻ ⇌ Cl ₂ (g) + 2 H ₂ O(l)	+ 1.61
MnO ₄ ⁻ (aq) + 8 H ⁺ (aq) + 5 e ⁻ ⇌ Mn ²⁺ (aq) + 4 H ₂ O(l)	+ 1.51
Au ³⁺ (aq) + 3 e ⁻ ⇌ Au(s)	+ 1.50
HClO(aq) + H ⁺ (aq) + 2 e ⁻ ⇌ Cl ⁻ (aq) + H ₂ O(l)	+ 1.48
PbO ₂ (s) + 4 H ⁺ (aq) + 2 e ⁻ ⇌ Pb ²⁺ (aq) + 2 H ₂ O(l)	+ 1.46
Cl ₂ (g) + 2 e ⁻ ⇌ 2 Cl ⁻ (aq)	+ 1.36
Cr ₂ O ₇ ²⁻ (aq) + 14 H ⁺ (aq) + 6 e ⁻ ⇌ 2 Cr ³⁺ (aq) + 7 H ₂ O(l)	+ 1.23
O ₂ (g) + 4 H ⁺ (aq) + 4 e ⁻ ⇌ 2 H ₂ O(l)	+ 1.23
MnO ₂ (s) + 4 H ⁺ (aq) + 2 e ⁻ ⇌ Mn ²⁺ (aq) + 2 H ₂ O(l)	+ 1.22
Br ₂ (l) + 2 e ⁻ ⇌ 2 Br ⁻ (aq)	+ 1.07
NO ₃ ⁻ (aq) + 4 H ⁺ (aq) + 3 e ⁻ ⇌ NO(g) + 2 H ₂ O(l)	+ 0.96
2 Hg ²⁺ (aq) + 2 e ⁻ ⇌ Hg ₂ ²⁺ (aq)	+ 0.91
Hg ₂ ²⁺ (aq) + 2 e ⁻ ⇌ 2 Hg(l)	+ 0.85
O ₂ (g) + 4 H ⁺ (aq) + 4 e ⁻ ⇌ 2 H ₂ O(l)	+ 0.82
NO ₃ ⁻ (aq) + 2 H ⁺ (aq) + e ⁻ ⇌ NO ₂ (g) + H ₂ O(l)	+ 0.80
Ag ⁺ (aq) + e ⁻ ⇌ Ag(s)	+ 0.80
Hg ₂ ²⁺ (aq) + 2 e ⁻ ⇌ 2 Hg(l)	+ 0.80
Fe ³⁺ (aq) + e ⁻ ⇌ Fe ²⁺ (aq)	+ 0.77
O ₂ (g) + 2 H ⁺ (aq) + 2 e ⁻ ⇌ H ₂ O ₂ (aq)	+ 0.68
MnO ₄ ⁻ (aq) + 2 H ₂ O(l) + 3 e ⁻ ⇌ MnO ₂ (s) + 4 OH ⁻ (aq)	+ 0.59
I ₂ (s) + 2 e ⁻ ⇌ 2 I ⁻ (aq)	+ 0.54
Cu ⁺ (aq) + e ⁻ ⇌ Cu(s)	+ 0.52
O ₂ (g) + 2 H ₂ O(l) + 4 e ⁻ ⇌ 4 OH ⁻ (aq)	+ 0.40
Cu ²⁺ (aq) + 2 e ⁻ ⇌ Cu(s)	+ 0.34
Cu ²⁺ (aq) + e ⁻ ⇌ Cu ⁺ (aq)	+ 0.16
Sn ⁴⁺ (aq) + 2 e ⁻ ⇌ Sn ²⁺ (aq)	+ 0.15
S(s) + 2 H ⁺ (aq) + 2 e ⁻ ⇌ H ₂ S(aq)	+ 0.14
2 H ⁺ (aq) + 2 e ⁻ ⇌ H ₂ (g)	0 exactly
Pb ²⁺ (aq) + 2 e ⁻ ⇌ Pb(s)	- 0.13
Sn ²⁺ (aq) + 2 e ⁻ ⇌ Sn(s)	- 0.14
Ni ²⁺ (aq) + 2 e ⁻ ⇌ Ni(s)	- 0.26
Co ²⁺ (aq) + 2 e ⁻ ⇌ Co(s)	- 0.28
PbSO ₄ (s) + 2 e ⁻ ⇌ Pb(s) + SO ₄ ²⁻ (aq)	- 0.36
Cd ²⁺ (aq) + 2 e ⁻ ⇌ Cd(s)	- 0.40
2 H ₂ O(l) + 2 e ⁻ ⇌ H ₂ (g) + 2 OH ⁻ (aq) [1.00 x 10 ⁻⁷ mol L ⁻¹]	- 0.41
2 CO ₂ (g) + 2 H ⁺ (aq) + 2 e ⁻ ⇌ HOOC-COOH(aq)	- 0.43
Fe ³⁺ (aq) + 2 e ⁻ ⇌ Fe(s)	- 0.44
Au(CN) ₂ ⁻ (aq) + e ⁻ ⇌ Au(s) + 2 CN ⁻ (aq)	- 0.60
Cr ³⁺ (aq) + 3 e ⁻ ⇌ Cr(s)	- 0.73
Zn ²⁺ (aq) + 2 e ⁻ ⇌ Zn(s)	- 0.76
2 H ₂ O(l) + 2 e ⁻ ⇌ H ₂ (g) + 2 OH ⁻ (aq)	- 0.83
Mn ²⁺ (aq) + 2 e ⁻ ⇌ Mn(s)	- 1.18
Al ³⁺ (aq) + 3 e ⁻ ⇌ Al(s)	- 1.66
Mg ²⁺ (aq) + 2 e ⁻ ⇌ Mg(s)	- 2.37
Na ⁺ (aq) + e ⁻ ⇌ Na(s)	- 2.71
Ca ²⁺ (aq) + 2 e ⁻ ⇌ Ca(s)	- 2.76
Sr ²⁺ (aq) + 2 e ⁻ ⇌ Sr(s)	- 2.89
Ba ²⁺ (aq) + 2 e ⁻ ⇌ Ba(s)	- 2.91
K ⁺ (aq) + e ⁻ ⇌ K(s)	- 2.93

6
C
12.01

← Atomic Number
← Symbol
← Atomic Weight

NAMES AND FORMULAE FOR SOME COMMON IONS AND POLYATOMIC IONS

+1 charge	-1 charge
lithium, Li^+ sodium, Na^+ potassium, K^+ copper (I) or cuprous, Cu^+ silver, Ag^+	fluoride, F^- chloride, Cl^- bromide, Br^- iodide, I^- hydride, H^-
ammonium, NH_4^+ hydrogen, $\text{H}_{(\text{aq})}^+$	hydroxide, OH^- nitrite, NO_2^- nitrate, NO_3^- acetate, CH_3COO^- hydrogencarbonate or bicarbonate, HCO_3^- hydrogensulfate or bisulfate, HSO_4^- chlorate, ClO_3^-
+2 charge	-2 charge
manganese Mn^{2+} magnesium, Mg^{2+} calcium, Ca^{2+} barium, Ba^{2+} zinc, Zn^{2+} copper (II) or cupric, Cu^{2+} mercury (II) or mercuric, Hg^{2+} iron (II) or ferrous, Fe^{2+} tin (II) or stannous, Sn^{2+} lead (II) or plumbous, Pb^{2+} nickel(II), Ni^{2+}	oxide, O^{2-} sulfide, S^{2-}
	carbonate, CO_3^{2-} sulfate, SO_4^{2-} sulfite, SO_3^{2-}

+3 charge	-3 charge
aluminium, Al^{3+} iron (III) or ferric, Fe^{3+} chromium, Cr^{3+}	nitride, N^{3-}
	phosphate, PO_4^{3-}
+4 charge	
tin (IV) or stannic, Sn^{4+} lead (IV) or plumbic, Pb^{4+}	

THESE NAMES AND FORMULAE MUST BE MEMORISED